

Customer Behavior Analysis

Exploratory data analysis of retail e-commerce transactions to drive business decisions



About the Dataset



Structured Tabular Data

Customer transaction-level retail dataset with 5,000 records



Mixed Data Types

Numerical (Age, Purchase Amount) and categorical (Category, Payment Method) fields



Business Purpose

Customer behavior analysis and strategic decision-making



Why This Data Matters



Strategic Benefits

- Identify high-value customers
- Understand purchase patterns
- Optimize discount strategies
- Improve marketing campaigns
- Increase customer lifetime value

Business Problem

How can retail companies understand customer purchasing behavior to increase revenue, improve retention, and optimize marketing?

Low Customer Retention

Customers not purchasing frequently enough

Ineffective Marketing

Discounts and promo codes may not drive purchases

Unclear Product Demand

Which categories and products perform best?

Payment Insights

Understanding preferred payment methods

Data Dictionary Highlights



Customer Profile

Customer ID, Age, Gender, Location, Subscription Status



Purchase Details

Item Purchased, Category, Purchase Amount, Season, Color, Size



Customer Feedback

Review Rating, Shipping Type, Payment Method



Discount Analysis

Discount Applied, Promo Code Used, Previous Purchases

Data Cleaning Process



Initial Exploration

Loaded data with pandas, checked structure, data types, and missing values



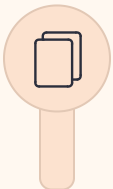
Category Correction

Fixed inconsistencies between Item Purchased and Category using mapping dictionary



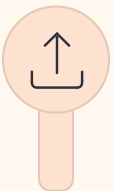
Missing Values

Applied category-specific logic: Electronics → "Not Applicable", Clothing → mode



Duplicate Removal

Identified and removed duplicates using Customer ID



Export to SQL

Standardized column names and exported cleaned data for analysis

Key SQL Findings

486K

Electronics Revenue

Top-performing category

219

Average w/ Discount

vs \$182 without

51%

Discount Usage

On laptops and phones

2.7K

Avg Spend: Subscribers

vs \$165 for non-subscribers



Power BI Dashboard Overview

Dashboard Highlights

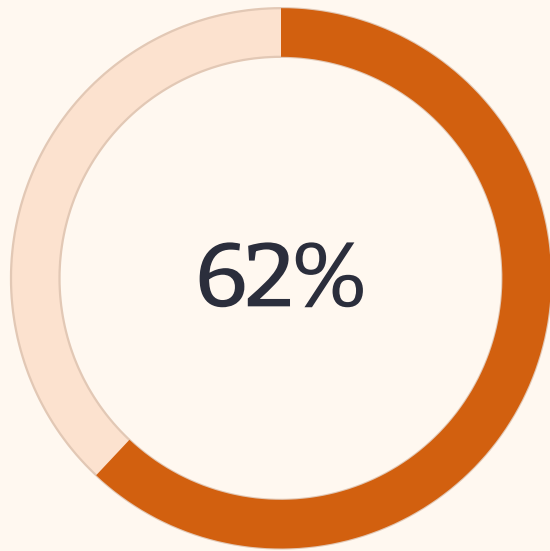
- 5,000 total customers
- 30 unique items
- \$994K total spend
- \$199 average purchase
- 3.62 average rating

Visualizations

- Revenue by gender and category
- Seasonal trends
- Age distribution
- Payment methods
- Shipping preferences

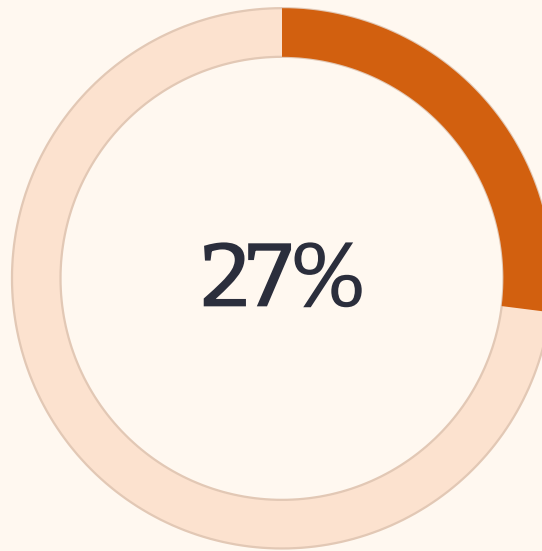


Customer Segmentation Insights



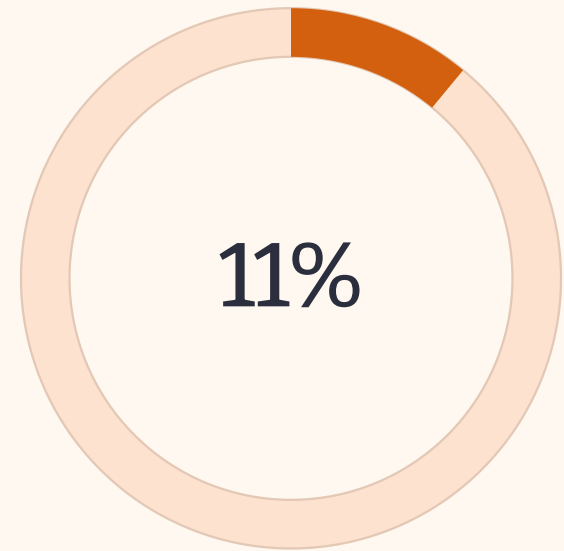
Loyal Customers

3,085 customers with high purchase frequency



Returning Customers

1,354 customers making repeat purchases



New Customers

561 first-time buyers

Strategic Recommendations



Focus on Electronics

Electronics drives 49% of revenue



Optimize Discounts

Discounts increase average purchase by 20%



Boost Subscriptions

Subscribers spend 64% more



Target 51+ Age Group

Generates 40% of total revenue

